

Two literary allegories

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Abstract

I believe strongly in trying to engage with literature. I would like to take two allegories from two books which I have read recently and which resonated with me in the context above. I hope that, even if you are more optimistic than me, you will get something out of these allegories, even if it is just the enjoyment of some writing and perhaps the desire to read more from these authors. I also hope that it can be a respite from reading directly about the philosophy and politics of mathematics.

Suggested tags: Reflection, Personal experience, Community

This article has been submitted to the Proofs and prompts blog. An appendix with some further discussion has been added to this version.

1 An apology of sorts

I am a doctoral student in something like geometric group theory. One could argue that my subfield (Kleinian groups, from a very constructive point of view) is close to dead, in the sense that all the big names in the field moved into other areas after the proof of various major conjectures of Lars Ahlfors, Albert Marden, and William Thurston dating back to the 60s and 70s, and I interact with my closest colleagues through journal articles published fifty years ago. I estimate there are less than five people worldwide who actually work with the same kinds of ideas as me closely day to day (most of whom are much more senior and coming to the midpoint or endpoint of their careers), and maybe about fifty who used to work in this area and are expert enough to review papers but who have moved on. My work has value because I still manage to find applications of this work to other areas of mathematics where there are active researchers who are genuinely interested in the results which I produce using this old technology—in September I put a preprint on the arXiv which connects some famous results in my field from the 1960s to ‘ q -deformed numbers’, which a lot of people seem to be interested in at the moment. I often feel like if I were not doing this work then from the point of view of broader mathematical

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society there would be essentially no difference: I am very confident that my proofs of theorems in other areas could be replaced with other proofs using other techniques which would provide just as satisfying of a ‘reason’ for those theorems to be true. Nevertheless, I feel like if I were to leave one day then my subfield would be one step closer to the day where the last person genuinely engages for an extended period of time with the set of ideas in my subfield beyond just a surface-level interaction where they skim theorem statements for applications.

Like many, but of course not all, of my colleagues of a similar career level who I have spoken to recently, I am now considering leaving mathematics to pursue a more stable career.

I do not want to describe in detail all the reasons for this, which have been covered at length in various other pieces of recent writing; some of the ones which are closest to my personal views are [1, 2, 4, 6], but I have also tried to deeply engage with many of the posts holding differing views and some of the ones I found most valuable (but unfortunately not personally reassuring) were [5, 9, 10, 11]. An LLM can certainly regurgitate all the ideas contained in the famous papers of the 60s and 70s in my subfield, but there will be one less person around who would actually be interested in engaging with these ideas deeply, maybe now there will be four people instead of five. Someone might still cite these ideas in their work to prove some ‘important’ conjecture, but it seems just as likely in the abstract that an LLM would come up with a proof of the same statement using some other mathematical technology—after all, to someone proving a theorem in a different field, having a proof passing via one dead subfield of complex analysis versus another is unimportant, and certainly when there are no researchers left in that dead subfield there is no *cultural* reason for them to choose one path over the other. I think broadly speaking if I were to leave, there would be no great loss to mathematics, and yet I would still be sad that I might be one of the last people to actually engage with these ideas for the sake of trying to understand them deeply for their own sake. This would be (I think) something of an intangible loss, but only a slight loss.

I am deeply uninterested in discussing whether it is good or bad to use LLMs for doing mathematical research for philosophical reasons of ‘purity’—I choose not to use them because I enjoy everything that I do and there is not much I would like to totally offload to a machine, and I have strong reservations (to put it lightly) about the social and political externalities of these tools and the companies which run them. I am also tired of the quantity of low-thought LLM generated text created without intent which is spreading everywhere on the internet and I refuse to add to the pile. After all, to a first approximation no-one reads my work in detail anyway, so why should I diminish my own enjoyment of my craft in order to produce even more text which no-one will read? I feel like I already produce enough externally applicable results to justify my work to the rest of mathematics.

As I tried to emphasise in my opening paragraph, I believe strongly in trying to engage with literature. I would like to take two allegories from two books which I have read recently and which resonated with me in the context above. I

hope that, even if you are more optimistic than me, you will get something out of these allegories, even if it is just the enjoyment of some writing and perhaps the desire to read more from these authors. I also hope that it can be a respite from reading directly about the philosophy and politics of mathematics.

2 Bohumil Hrabal, *Too Loud a Solitude*

I didn't pick up the books that had fallen at my feet, I shoveled them straight into the drum, and even though I couldn't help noticing—my heart broke when I saw it—the Metaphysics of Morals, in it went with the rubbish, and on and on I worked, making bale after bale, no Old Masters this time around, a bale is a bale is a bale, I did only what I was paid for, my artistic days were over, and I realized that if I did only what I was supposed to do, I could be a one-man Brigade of Socialist Labor, and if I increased my output by fifty percent I could have a chance at the factory chalet in the mountains and, more important, a holiday in fabulous Greece—in other words, a chance to put on my long Johns and run around the track at Olympia and pay my respects to Aristotle in Stagira. So I kept slurping milk and working, working inhumanly, unfeelingly, the way they worked at the gigantic press in Bubny, and in the evening, when I had finished and proved I wasn't such a nitwit after all, my boss, who was having a shower in the facilities behind the office, yelled out to me from under the spray that he wasn't going to waste any more time on me and had sent a letter to the higher-ups telling them to do with me as they saw fit. [8, p. 73]

3 Mircea Cărtărescu, *Solenoid*

By day she was still the chemistry teacher, envied by all her colleagues for her clothes and shoes and purses, her house with 156 panes of glass, and her ministry husband, but by night, two or three times a week, dressed in black without makeup or perfume, in a headscarf and shoes the janitors wore, with tears dancing in her eyes and a dark hatred over her face like a dead god of love, and holding a crude sign made from a television box proclaiming, “Down with Aging!”, she stole out of her house and went to the Saint Elefterie church, where she joined a pack of somber silhouettes, all coming from the neighborhood Cotroceni, then walked through the quiet of the ruined city, haunted by nostalgia and old trams, toward the cemeteries: Ghencea, Bellu, Andronache, Izvorul Nou, Armenesc, Străulești, Adormirea Maicii Domnului, Metalurgiei, Israelit, Eternitatea, Colentina, Berceni, Luteran, Progresul, Saint Vineri. Through the narrow streets between abandoned, empty buildings

with their doors flung open, rotted like old barrel staves, with plaster allegorical statues spreading their flaking wings under the full moon, their group met up with groups from other neighborhoods, other people with shadowy faces and shining eyes. At the cemetery gates, beyond whose tall fence, profiled against the still vaguely illuminated sky, rose the stubby cupolas of the crypts and stumps of pitch-black crosses, the Picketists, sometimes a few dozen people, sometimes hundreds, took out their signs (“Down with Death!”, “Down with Illness!”, “Down with Agony!”, “Down with Suffering!”, “Protest Pain!”, “Pro Eternal Life!”, “Pro Eternal Consciousness!”, “Pro Human Dignity!” “NO to Passivity!”, “NO to Laziness!”, “NO to Resignation!”), and they began to silently pace in circles, for hours, in the cold of the night, with the determination of those who, as they say, “won’t go down without a fight.” “We’re always demonstrating, we’re always fighting against the terrible things people do, in their meanness and blindness, but how can we accept the destruction of the spirit, the enormous absurdity of its life and death in flesh? How can you await your execution, as silent as sheep and just as innocent, when you are a spirit, when you are a part of the Godhead? We, the Picketists, we scream silently against this unbelievable, unqualifiable, unpardonable human genocide. We hold that it must end. As free and dignified people, we must protest fate and fatalism. We do not accept them, we will not bow down to fate, and if we must perish, at least we will know it was not without revolt, without a yell, without indignation. We are the only ones who will die on our feet, not on our guts in the dirt, or on our knees.” Caty had heard these words dozens of times. Once every cemetery in Bucharest had been picketed for nights on end, the members of the sect, all in black and protesting before the blind and illiterate dead with crooked signs, moved to the crematorium, hospitals, the state militia, and the Securitate. They were arrested in droves, committed to psychiatric hospitals, locked up with political prisoners in miserable jails, but those who remained were always seen, with their signs, with their eyes bathed in tears, wherever the carnivorous flower of human suffering appeared. As though they had an intuition for pain and misfortune, they would arrive at a fire before the firemen and to serious car accidents before the ambulances. More than once, Caty came home bruised by the rubber batons of patrolmen or drenched by water cannons. But she found, in the end, her peace. In the end, she no longer feared getting older and disappearing. She fought against them, she hoped against them. The odds of winning weren’t important, but the fight was. Caty had separated herself from the lazy mass of hostages who lived with bowed heads. [3, p. 144–146]

4 Discussion

I apologise for giving something of a surface-level discussion of these passages; both of these books are complex pieces of art and it would be very easy to write a much longer discussion of each, but this is not the place for that.

The protagonist, Haňt'a, of Hrabal's *Too Loud a Solitude* is living in 1970s Prague. He works alone, crushing books into pellets to be recycled, destroying banned knowledge. He rebels by reading the books he is meant to be crushing and by taking pride in his work even though it is ultimately destructive. He is replaced by new workers who take (from his perspective) no pride in their work, viewing it only as a thing which they are told to do for eight hours a day so that they can be paid. One could argue that the work itself has little value to society (destroying books in order to produce slightly cheaper paper), and that the only value came from its effect on Haňt'a as an individual; when the process is automated, there is still minimal to society and the value to the individual is gone. Haňt'a spends twenty pages coming to terms with this, and then steps into his hydraulic press and crushes himself with his remaining books, at peace. To be clear, one should take this allegorically: to perhaps be a bit too on-the-nose about it, his decision is akin to a person finding they have been automated out of a job that was their way of life, and so leaving everything behind to start an entirely new life.

Compare this to Caty's futile protests against death in the 'magic realism' version of Bucharest: we all know that we are to die, we all know that it is pointless to bargain with death and the best we can do is to try to prolong life with medical care and healthy living, and yet the Picketists seem to derive tremendous meaning and optimism from fighting an unwinnable and (from our point of view) entirely futile protest against the concept of death. Compared to *Too Loud a Solitude*, there is a much less clear ending to *Solenoid*; somehow the protests succeed in changing things, but not in the sense of ending death—rather, they cause the decaying city around them to lift off into the sky, leaving all the citizens of Bucharest lost, homeless, bewildered, but together and ready to rebuild something. Even working together as a group, the Picketists are just as ineffective as Haňt'a at halting an inevitable end, but somehow they manage to create an end of a different kind than simply ceasing to exist.

I find both of these books beautiful in their own way. Hrabal, even in translation, is (in my opinion) one of the greatest prose writers of the last sixty years, and I find Cărtărescu's writing often deeply disturbing in a way that would be impossible if he did not deeply understand the human psyche. Reading both of them helped me come to terms in some way with what is going on in my career and in society at large; as a person who works largely alone with old literature rather than by discussing work with people I identify strongly with Haňt'a in feeling hopeless that my way of working is becoming less and less tenable, and exhausted and ready to give up. At the same time I feel like it is important to stand up with my peers like Caty and say what I believe, even if many people think I am insane to be trying to stop something so big and inevitable.

I do not have any resolution to this tension between these two points of view; like many people I am stressed and confused and have no answers. Most of all, I feel very tired. I just hope that you can take something out of these pieces of writing and perhaps be more successful than me in deciding for yourself.

I am glad that I learned arithmetic at school—how else could I experience its perils and pleasures?—Janet Frame [7]

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Appendix A Some further remarks

I tried, in the main body of this article, to stay away from ‘philosophy of the mathematical profession’ and to just speak about my experience in my subfield. Here I collect some more explicit things just because I want to write them down to clarify my own thinking. Please do not take the fact that I make this list public as an invitation to debate, you should form your own opinions and feel free to either keep them to yourself or publish them in some passive medium like a blog (or a parchment in a bottle thrown into a river).

- I am worried about the long-term impact of significant LLM usage on the brain. Certainly there is a lot of anecdotal evidence of people with pre-existing psychological conditions becoming psychotic when relying on LLMs as counsellors, therapists, or replacements for other human conditions; and there also seems to be evidence of addiction to LLM usage, which is arguably in the financial interests of the companies running them. In my view the clinical impacts on the ‘average’ brain will only become clear over an extended period of time. It would not surprise me if the long-term impacts would be similar to extended usage of some mild drug, where function is essentially unimpaired in the short term but there is degeneration of some kind in the long term.¹
- A lot of the panic among young people seems to actually be caused by the large number (not a majority, but a significant amount) of senior mathematicians who first spent a year or two hyping up this technology and encouraging the participation of major companies and who now (once it became clear to them that it is not in the best interest of these companies to encourage mathematical research) are loudly discussing how publishing and hiring practices must change immediately and how everyone needs to change the way they fundamentally do research *right now*. Many people have sacrificed quite a bit, either in real terms or in potential earnings/relationships/social life (by moving long distances to take a low-paying position in order to do mathematics). When senior people suddenly start saying that everything must change right now without taking a sufficiently measured approach or showing clear leadership then this causes people without an established career to panic.

¹One much circulated preprint is Nataliya Kosmyna et. al., “Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task”, arXiv: 2506.08872, but this does not seem to have been peer-reviewed. There is also: Hamsa Bastani et. al., “Generative AI without guardrails can harm learning: Evidence from high school mathematics”, PNAS 122 (2025), doi: 10.1073/pnas.2422633122; Yuhe Ke et. al., “AI-induced never-skilling in medical education”, Nature Medicine 32 (2026), doi: 10.1038/s41591-026-04438-y. I encourage mathematicians to check on PubPeer before heavily relying on these studies, as I found at least one paper referenced by the Nature Medicine paper which appears to have significant problems, <https://pubpeer.com/publications/6FC2E572462EEAC7A766E3322F6098>. I also want to point out that there is a significant recent body of work on cognitive and technical debt in software development due to LLM usage and many of the same issues appear to be applicable to mathematical use, but there is no paper in particular that I want to highlight here.

- I feel that the small but vocal set of people who believe that future mathematical research *must* be entirely LLM-driven is particularly alarming. Again, many junior people without a permanent position have made a lot of hard decisions based on the fact that they enjoy doing mathematics. If the job turns into simply prompting an LLM and writing up the results, then why should they subject themselves to the incredibly harsh academic job market, when this is exactly the same job as much of the software industry but with more cutthroat competition, lower pay, and far less long-term stability? I do not have any idea whether they are right (though I hope not), but I believe that their rhetoric is unnecessarily disruptive and stressful to young people who already need to deal with navigating a hostile job market and a totally broken publication system where it takes years to get publications accepted (or years to get rejected!). People should carefully think about how their opinions come across when they speak from a position of authority, and if they truly believe these things then they should be providing appropriate support to young people beyond simply telling them that this is the future and we should learn to deal with it. For instance, giving practical advice about what they think it is best to do in order to pivot to such a world—not just ‘start writing papers with LLMs’, rather (i) being very clear about what sorts of results they think will or will not have value, (ii) what their expectations will be for PhD students in terms of knowledge or research ability, (iii) what role junior positions (e.g. postdocs) will have in their own research groups.
- I am a bit disappointed in the number of people who also feel like they need to run every little piece of writing through an LLM to check grammar. Firstly, if you keep feeding your text through the model it is always going to be able to find something minor to change regardless of whether it is actually necessary, just because of how the technology works. Secondly, I think you should be proud of your literary voice.
- Finally, in principle I am not against tools like LLMs being used to explore ideas provided they are used to create text ‘with intent’ and provided that it is understood that generating a proof (or other forms of research like searching for literature) means knowing an answer to a question but missing out on all interesting side-paths and incidental knowledge which would have come from doing research in a more frictioned environment. However, with the current highly centralised LLMs asking for ‘new’ theorems without a very specific direction has a significant ethical problem: it cannot be guaranteed that the result is not distilled from an unpublished draft of some other researcher (possibly very junior or otherwise vulnerable). My point of view on this is currently taken from the sciences. There, one must generate a hypothesis before conducting the experiment in order to avoid finding statistically significant results by chance². Here, I feel that

²<https://xkcd.com/882/>

one should at the very minimum have a conjecture that a theorem of the desired form actually exists before trying to look for it to reduce the statistical chance that the model is just spitting something out from its training data. (Again this is something I feel uncomfortable about even with such guidelines but I already discussed this in detail in the body).

- I think that *Solenoid* rewards closer reading. I gave a very surface level interpretation above—fundamentally I read the book as being about the fundamental psychological need for humans to fight against the decay of the body (symbolised by the city of Bucharest), even though it is inevitable, and the importance of literature and broader human-made culture in this fight. I would welcome discussion of this book much more than discussion of mathematical philosophy or mathematical fortune telling. Hrabal's book is a lot shorter and (I think) much less opaque; one can find a few discussions of it in English in various books.